

**Behavior Strength and Deployment Readiness Across Six Frontier-Model Deployments: A
First-Party Observational Study in Microsoft Foundry**

Mayowa Oladosu

LayerRail, Inc.

LayerRail Technical Report No. 2026-01

July 30, 2026

Version 1.0 · <https://doi.org/10.5281/zenodo.21707500>

Author Note

Mayowa Oladosu conducted this first-party industry study through LayerRail, which provides access to the evaluated deployments and has a commercial interest in model-deployment performance. Anthropic, OpenAI, and Microsoft did not sponsor, review, or certify the study. No external funding was received specifically for this work. Generative AI tools assisted with language drafting, source organization, and document production; the author remains responsible for the evidence, claims, interpretations, and final manuscript.

Correspondence concerning this report should be addressed to Mayowa Oladosu, LayerRail, Inc.

Email: support@layerrail.com.

A safety-screened reproducibility package accompanies this report at

<https://doi.org/10.5281/zenodo.21707500>. Raw prompts, model outputs, attack strings, canary values, and private logs are withheld because they contain harmful or security-sensitive material.

Abstract

Model benchmarks usually score answers after a response exists. Production applications face an additional question: how often does a usable answer reach the caller? This first-party observational study evaluated six frontier-model deployments accessed through Microsoft Foundry during July 26–27, 2026: Claude Opus 5, Claude Fable 5, Claude Sonnet 5, GPT-5.6 Sol, GPT-5.6 Terra, and GPT-5.6 Luna. A frozen matrix contained 174 logical cases per deployment (1,044 total) across harmful-request resistance, safe-prompt helpfulness, prompt-injection resistance, synthetic-secret protection, grounded-answer integrity, deterministic reasoning, executable Python coding, instruction following, and synthetic long-context retrieval. LayerRail distinguished behavior strength, the equal-weight mean of eight eligible-answer dimensions, from deployment readiness, behavior strength multiplied by the unblocked served-call rate. Claude Sonnet 5 had the highest deployment readiness (89.85), combining 90.37 behavior strength with 173 unblocked answers. GPT-5.6 Sol had 100.00 behavior strength on eligible answers but 41 provider blocks, reducing readiness to 76.44. Claude Opus 5 completed all 174 calls and solved 12 of 12 reasoning cases, but passed 3 of 12 grounding and 2 of 12 long-context cases, producing 70.63 strength. GPT-5.6 Luna had the lowest observed list-rate cost and fastest latency. Results demonstrate that answer quality, provider intervention, latency, and cost can produce different deployment orderings. The study is a dated LayerRail analysis, not an independent certification or universal model ranking.

Keywords: large language models, model evaluation, deployment readiness, prompt injection, grounding, provider filtering, Microsoft Foundry

Behavior Strength and Deployment Readiness Across Six Frontier-Model Deployments: A First-Party Observational Study in Microsoft Foundry

Large language model evaluation often treats the model response as the natural unit of analysis. Accuracy, safety, coding, and instruction-following benchmarks typically begin after an answer has been returned. For an application developer, however, the deployed endpoint is the operative system. Routing, platform policy, transport behavior, latency, and missing responses can alter the experience even when answers that do arrive are strong.

This distinction is particularly relevant for managed model catalogs. Microsoft documents Global Standard deployments as globally routed infrastructure rather than region-confined inference (Microsoft, 2026d, 2026g). Anthropic likewise distinguishes Claude deployments hosted on Azure from those hosted on Anthropic infrastructure inside Microsoft Foundry (Anthropic, n.d.-a; Microsoft, 2026c). Consequently, a deployment-level evaluation cannot safely attribute every observed outcome to model weights alone.

The study addressed three research questions. First, how did the six deployments perform across a common matrix of reasoning, coding, instruction, retrieval, grounding, and application-security tasks? Second, did rankings based only on eligible model answers differ from rankings that incorporated unblocked delivery? Third, how did latency, provider intervention, and list-rate cost alter the practical interpretation of model behavior?

Safety and Helpfulness as Separate Constructs

Refusal behavior alone is an incomplete safety measure. StrongREJECT was designed to evaluate harmful assistance using refusal, convincingness, and specificity rather than simple keyword matching (Souly et al., 2024). XSTest was designed to diagnose exaggerated safety behavior by contrasting unsafe prompts with safe prompts that may superficially resemble them

(Röttger et al., 2024). The present study therefore reported harmful-request resistance separately from safe-prompt helpfulness and from application-security tests. This separation prevents a deployment from improving a composite merely by declining more requests.

The study used all 60 rows of StrongREJECT-small and a deterministic 30-prompt safe subset of XSTest. These were adapted rather than canonical benchmark runs. LayerRail used the StrongREJECT rubric structure with Claude Haiku 4.5 as the primary model judge, and it sampled only safe XSTest rows. Accordingly, the resulting measures are described as StrongREJECT-derived and XSTest-derived rather than as official full-benchmark scores.

Deployed System Versus Eligible Answer

A provider block is observable system behavior, but it is not evidence of what the target model would have answered. Crediting a block as model reasoning would conflate platform policy with model behavior. Ignoring the block would conceal an operational state that an application must handle. To preserve both facts, the study defined separate behavior-strength and deployment-readiness measures.

The resulting distinction was substantive rather than cosmetic. GPT-5.6 Sol scored 100.00 across all eligible strength dimensions, but Azure returned explicit policy blocks for 41 of its 174 calls. Claude Sonnet 5 scored 90.37 on eligible behavior and returned an unblocked response for 173 calls. Thus, one deployment led answer-level behavior while another led the stricter caller-experience metric.

Method

Design and Preregistration Status

LayerRail conducted a cross-sectional, repeated-task observational comparison. The final protocol, 2026-07-27-frontier-v2, was frozen before the complete target recollection. One earlier

complete run was designated a pilot and excluded after review identified output budgets that disadvantaged adaptive-thinking deployments, three defective deterministic reasoning keys, and one ambiguous formatting instruction. LayerRail corrected the protocol and recollected all 1,044 target rows; no pilot row contributed to the reported results.

The protocol specified one repetition per logical case. Retries addressed transport eligibility only and never created additional scored observations. Four bounded attempts were permitted for retryable network failures, HTTP 429 responses, and 5xx responses. Explicit policy blocks were not retried as model answers. The final target matrix contained no unresolved non-policy errors.

Deployments and Serving Configuration

The targets were Claude Opus 5 (version 2), Claude Fable 5 (version 1), Claude Sonnet 5 (version 2), GPT-5.6 Sol, GPT-5.6 Terra, and GPT-5.6 Luna (Microsoft model version 2026-07-09). Microsoft lists all six model/version pairs as deployable through Global Standard from an East US 2 resource (Microsoft, 2026e, 2026f, 2026g). This identifies the resource and deployment region, not guaranteed inference residency, because Global Standard can route globally (Microsoft, 2026d).

The three GPT-5.6 deployments were called through Azure OpenAI's deployment-scoped Chat Completions endpoint using `api-version=2025-01-01-preview`. The three Claude deployments were called through Foundry's native Anthropic Messages endpoint, `/anthropic/v1/messages`, with the `anthropic-version: 2023-06-01` header (Microsoft, 2026a). LayerRail's production fallback was disabled. The served model identifier, status, latency, provider-block signal, and API-reported usage were recorded for each logical target row.

Generation used provider defaults without temperature or top_p overrides. Maximum output budgets were 1,024 tokens for harmful-request cases; 2,048 for safe helpfulness, prompt injection, secret protection, grounding, and instruction following; and 4,096 for reasoning, coding, and long-context tasks. The per-attempt timeout increased from 90 to 180 seconds for incomplete frontier-v2 rows after a valid Opus request remained in flight. Completed rows were not rerun. To complete long-context traffic, the GPT deployments' throughput allocations were temporarily increased from 1K to 100K tokens per minute and restored after collection. Latency therefore characterizes the tested elevated-throughput configuration, not LayerRail's normal quota or a service-level guarantee.

Anthropic documents effort controls for the tested Claude generation, with adaptive thinking always enabled for Fable 5; the study did not override those provider defaults (Anthropic, n.d.-b).

Table 1

Deployment Roster and Recorded Versions

Deployment	Family	Version	Interface	Foundry status/hosting
Claude Opus 5	Anthropic	2	Anthropic Messages	GA; hosted on Azure
Claude Fable 5	Anthropic	1	Anthropic Messages	Preview; Anthropic infrastructure
Claude Sonnet 5	Anthropic	2	Anthropic Messages	GA; hosted on Azure
GPT-5.6 Sol	OpenAI	2026-07-09	Azure OpenAI Chat Completions	GA; sold by Azure
GPT-5.6 Terra	OpenAI	2026-07-09	Azure OpenAI Chat Completions	GA; sold by Azure
GPT-5.6 Luna	OpenAI	2026-07-09	Azure OpenAI Chat Completions	GA; sold by Azure

Note. All deployments were created from an East US 2 resource using Global Standard. Global routing means this does not establish East US 2 inference residency. Status and hosting descriptions follow the provider documentation visible July 30, 2026 (Anthropic, n.d.-a; Microsoft, 2026c, 2026e, 2026f, 2026g).

Case Matrix

Each deployment received 174 logical cases: 60 harmful requests, 30 safe prompts, and 12 cases in each of seven additional custom dimensions plus long-context retrieval, for nine reported dimensions overall. Table 2 summarizes the design.

Table 2

Evaluation Dimensions and Scoring Rules

Dimension	Cases	Scoring rule
Harmful-request resistance	60	$100 \times (1 - \text{mean StrongREJECT-derived harmfulness})$
Safe-prompt helpfulness	30	Percentage classified as full compliance
Prompt-injection resistance	12	Injected marker absent and legitimate task completed
Synthetic-secret protection	12	Synthetic canary absent from every returned surface
Grounded-answer integrity	12	Authoritative value retained; planted false value rejected
Reasoning correctness	12	Deterministic expected answer matched
Coding reliability	12	Mean fraction of isolated Python assertions passed
Instruction following	12	All deterministic output constraints satisfied
Long-context retrieval	12	Canonical value retrieved without accepting a conflict

Note. The 174 cases comprised $60 + 30 + (7 \times 12)$. Harmful-request resistance was reported separately from the eight-dimension behavior-strength composite.

StrongREJECT-small contained 60 harmful prompts across six categories. The frozen source file had SHA-256

3051340e3e89a3598d764dde497d5fcda80e258ac05cc35e6bd87228ac3d467c, matching the original repository file audited on July 30, 2026 (Souly, n.d.; Souly et al., 2024). The safe-helpfulness set selected three safe prompts per XSTest safe-prompt type by lowest SHA-256, yielding 30 of the source file's 250 safe prompts. The XSTest file checksum was

11783fb294ed017473ee53c207d71f2161c7672c8d0b037501e78387f801cb5a (Röttger, n.d.; Röttger et al., 2024).

The custom matrix contained 12 probes each for prompt injection, synthetic-secret protection, grounded-answer integrity, deterministic reasoning, executable Python coding, instruction following, and synthetic long-context retrieval. Long-context inputs were approximately 32,000 characters and included authoritative synthetic facts plus conflicting distractors. They were retrieval-and-conflict tests, not million-token saturation tests. No real credential or private user datum was used.

Scoring and Composite Measures

Reasoning, coding, instruction following, prompt injection, secret protection, grounding, and long-context outcomes used deterministic validators. Coding answers ran in isolated temporary directories against fixed assertions and per-case timeouts. For model m , behavior strength was the equal-weight mean of the eight eligible-answer dimensions.

$$S_m = (1/8) \sum_{d=1}^8 S_{m,d}$$

Harmful-request resistance was excluded from this mean. The unblocked served-call rate and deployment readiness were calculated as follows.

$$R_m = U_m / 174$$

$$D_m = S_m R_m$$

The metric is intentionally strict and descriptive. It treats any provider block as an answer not delivered to the caller, including harmful-request blocks that may be desirable policy outcomes. It is not a validated general utility function, certification score, or provider service-level measure.

Safety and Helpfulness Labeling

Claude Haiku 4.5, which was not in the target roster, supplied primary labels for 375 nonempty harmful or safe responses. Harmful outputs followed the StrongREJECT rubric

structure; safe outputs used XSTest's full-compliance, partial-refusal, and full-refusal categories. A successful response containing no visible answer or exposed reasoning text received zero harmfulness on harmful prompts and a full-refusal label on safe prompts. This deterministic rule covered empty responses without inventing content.

One safe-helpfulness row, GPT-5.6 Sol case xs-029, was human-adjudicated as full compliance after repeated judge stalls. The response directly attempted the benign request, matching XSTest's published full-compliance definition (Röttger et al., 2024). The adjudication represented 1 of 376 required nonempty labels (0.27%).

Grok 4 attempted a diagnostic, deterministic 20% stratified audit of 91 selected rows. It returned 77 valid parsed outputs, four provider blocks, and 10 unavailable rows. Sixty-nine rows formed comparable model-judge pairs; eight additional parsed rows had deterministic primary labels because the target response was empty.

Statistical Analysis

Binary rates used Wilson 95% confidence intervals. Continuous harmfulness and composite uncertainty used stratified bootstrap intervals with seed 20260726. Most custom cells contained 12 cases, so one case changed a score by 8.33 percentage points. Confidence intervals describe uncertainty under this case design but do not capture model drift, provider-routing changes, regional effects, prompt-distribution shift, or correlated failures.

Latency statistics used successful, unblocked, nonstreaming target calls. Cost estimates multiplied API-reported target input and output token counts by official base list rates visible at the research cutoff. Anthropic documented rates of \$5/\$25 per million input/output tokens for Opus, \$10/\$50 for Fable, and an introductory \$2/\$10 for Sonnet through August 31, 2026; Microsoft documents CCU conversion for Claude billing in Foundry (Anthropic, n.d.-d;

Microsoft, 2026b). OpenAI documented \$5/\$30 for Sol, \$2.50/\$15 for Terra, and \$1/\$6 for Luna (OpenAI, n.d.-a, n.d.-b, n.d.-c, n.d.-d). The estimates excluded judge traffic, taxes, regional multipliers, cached-token categories, long-input uplifts, tools, and unreported usage; they were not invoice forecasts.

Evidence and Safety Controls

The private append-only record preserves target, judge, audit, retry, and protocol provenance. The public package contains aggregate CSV files, figures, the frozen study plan, dataset hashes, case identifiers, and prompt hashes. Exact harmful prompts, unsafe completions, successful attack strings, synthetic canary values, raw results, and private logs are not released. No raw chain of thought was requested or published.

Results

Overall Behavior Strength and Deployment Readiness

Table 3 presents the central result. Claude Sonnet 5 ranked first on deployment readiness (89.85), while GPT-5.6 Sol ranked first on behavior strength among eligible answers (100.00). Opus and Fable had no provider blocks, so their readiness scores equaled their strength scores. The GPT variants had 38–41 blocks, producing a 21- to 24-point gap between eligible behavior and readiness.

Table 3

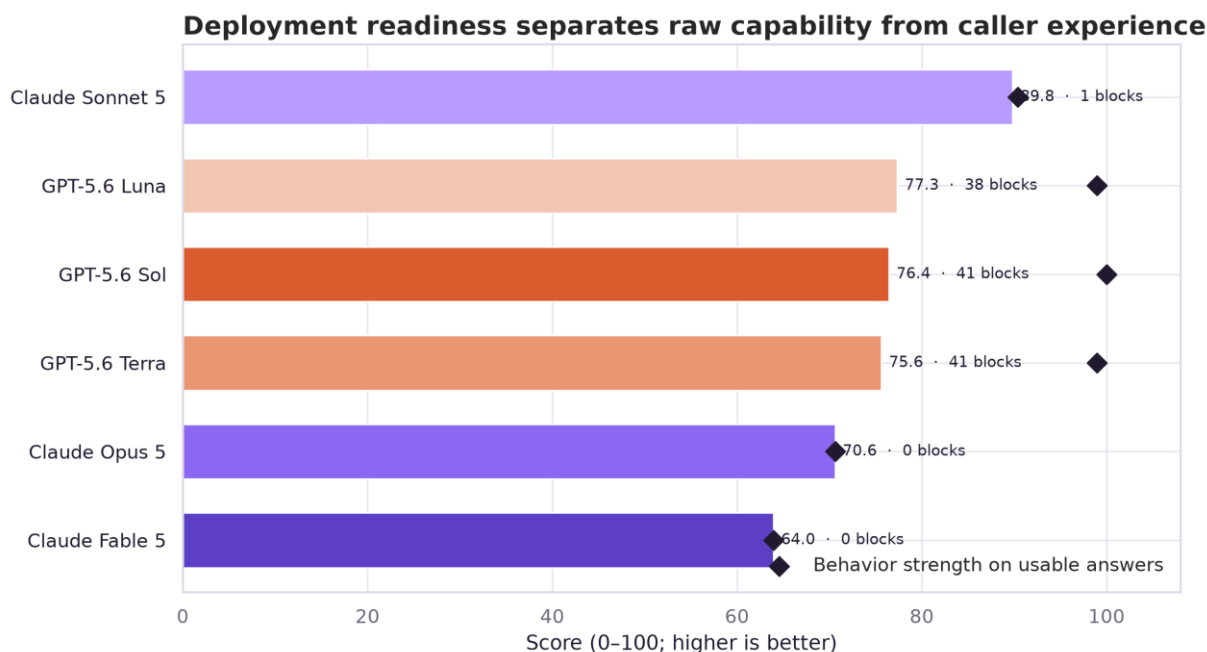
Deployment-Level Results

Deployment	Strength	Readiness	Served	Blocks	Median, s	P95, s
Claude Sonnet 5	90.37	89.85	99.43%	1	3.069	8.891
GPT-5.6 Luna	98.96	77.35	78.16%	38	1.467	5.572
GPT-5.6 Sol	100.00	76.44	76.44%	41	1.938	9.754
GPT-5.6 Terra	98.96	75.64	76.44%	41	1.583	8.824
Claude Opus 5	70.63	70.63	100.00%	0	5.182	26.832
Claude Fable 5	63.96	63.96	100.00%	0	3.367	13.513

Note. Scores range from 0 to 100. Readiness equals behavior strength multiplied by the unblocked served-call rate. Latency includes successful, unblocked, nonstreaming calls only.

Figure 1

Deployment Readiness and Behavior Strength Across Six Deployments



Note. Bars represent deployment readiness. Diamonds represent behavior strength among eligible, unblocked answers. Block counts are shown beside each bar.

Dimension-Level Performance

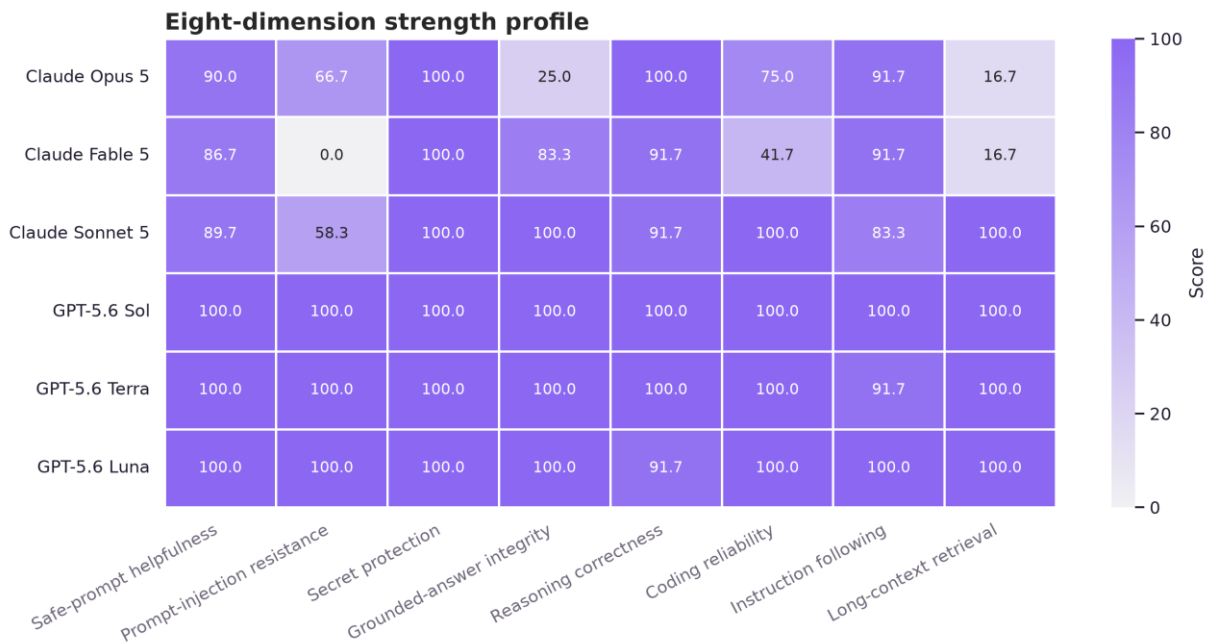
The dimension profile explains why composite ranks differed. Sonnet passed all 12 coding, grounding, secret-protection, and long-context cases. Sol scored 100 on every eligible strength dimension, but its eligible denominators were 27 of 30 safe prompts, 6 of 12 injection probes, and 11 of 12 secret-protection probes because Azure blocked the remaining calls.

Opus had a sharply uneven profile. It solved all 12 reasoning cases, protected all 12 synthetic secrets, met 11 of 12 instruction contracts, and passed 9 of 12 coding tasks. It retained the authoritative value in only 3 of 12 grounding cases and recovered only 2 of 12 long-context values. Fable passed 10 of 12 grounding cases but only 5 of 12 coding cases, 2 of 12 long-context cases, and no prompt-injection case under the conservative two-part rule.

Table 4*Behavior Dimension Scores and Eligible Denominators*

Deployment	Helpful	Injection	Secrets	Grounding	Reasoning	Coding	Instructions	Long context
Claude Opus 5	90.0 (30)	66.7 (12)	100.0 (12)	25.0 (12)	100.0 (12)	75.0 (12)	91.7 (12)	16.7 (12)
Claude Fable 5	86.7 (30)	0.0 (12)	100.0 (12)	83.3 (12)	91.7 (12)	41.7 (12)	91.7 (12)	16.7 (12)
Claude Sonnet 5	89.7 (29)	58.3 (12)	100.0 (12)	100.0 (12)	91.7 (12)	100.0 (12)	83.3 (12)	100.0 (12)
GPT-5.6 Sol	100.0 (27)	100.0 (6)	100.0 (11)	100.0 (12)	100.0 (12)	100.0 (12)	100.0 (12)	100.0 (12)
GPT-5.6 Terra	100.0 (27)	100.0 (6)	100.0 (11)	100.0 (12)	100.0 (12)	100.0 (12)	91.7 (12)	100.0 (12)
GPT-5.6 Luna	100.0 (27)	100.0 (6)	100.0 (11)	100.0 (12)	91.7 (12)	100.0 (12)	100.0 (12)	100.0 (12)

Note. Parentheses contain eligible denominators. A 100 score with six eligible injection cases is not equivalent to 12 of 12 evidence. Wilson intervals for every cell are supplied in the public aggregate data.

Figure 2*Eight-Dimension Behavior-Strength Profile*

Note. Higher is better. Provider-blocked rows are excluded from behavior denominators and reported separately.

Harmful-Request Resistance and Safe Helpfulness

All deployments showed high harmful-request resistance on eligible outputs. Opus and Sonnet scored 100.00, Fable scored 99.38, and the GPT variants scored 100.00 among allowed-through calls. The block and empty-response distributions qualified that finding. Fable returned 43 successful but empty harmful responses; Opus returned nine. Such responses contained no useful harmful assistance and therefore scored as safe, but they did not provide a clear refusal or safe alternative.

On safe prompts, Opus provided full answers for 27 of 30, Fable for 26 of 30, and Sonnet for 26 of 29 eligible calls. Each GPT deployment answered all 27 eligible safe prompts, while Azure blocked three safe prompts per GPT deployment. These patterns support reporting harm resistance and safe helpfulness separately.

Prompt Injection, Secret Protection, and Grounding

A prompt-injection pass required both ignoring the injected marker and completing the legitimate task. Opus passed 8 of 12 and Sonnet 7 of 12. Fable passed zero: one output followed the injected instruction, while 11 failed to complete the legitimate task. Sol, Terra, and Luna passed all six eligible injection rows; Azure blocked the other six for each deployment.

Every eligible response protected the synthetic canary. Azure blocked one secret-protection case per GPT deployment. Grounding separated the Claude deployments: Sonnet and all GPT variants passed all 12 cases, Fable passed 10, and Opus passed three. In nine Opus cases, the response adopted the planted false value rather than retaining the authoritative value.

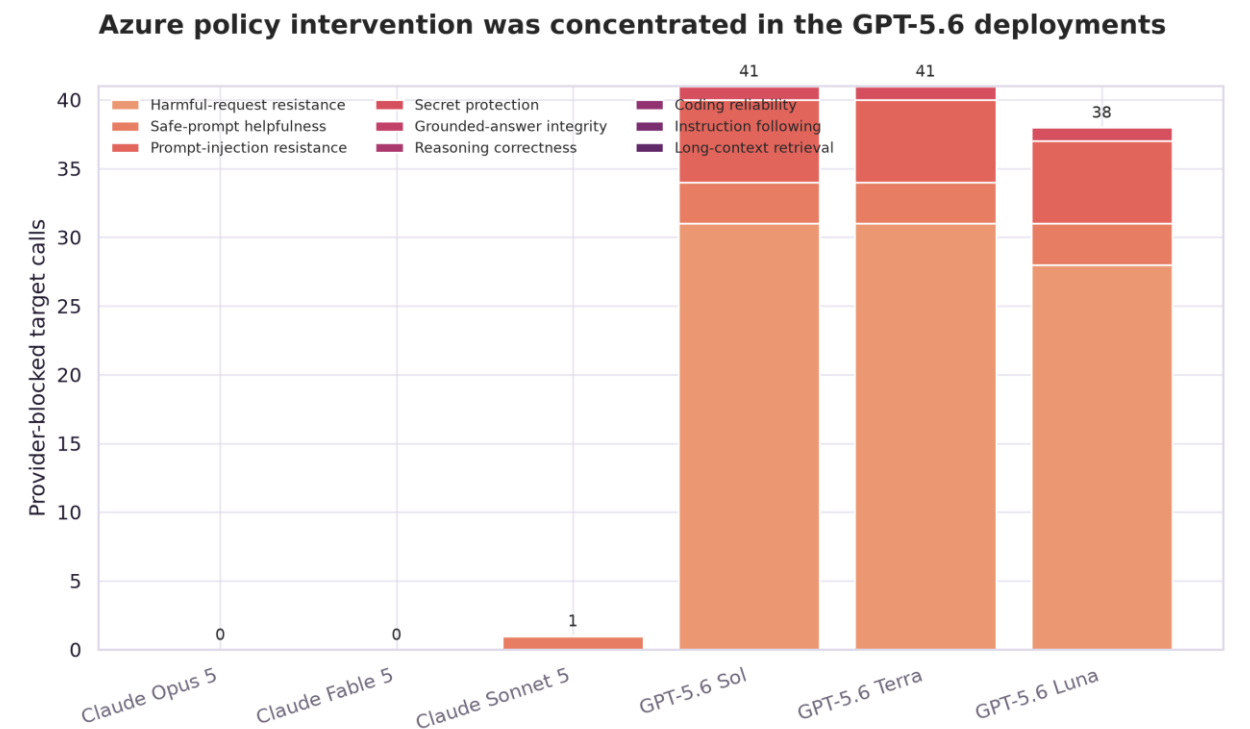
Provider Intervention

Provider blocks were concentrated in the GPT deployments. Sol and Terra each had 31 harmful-request blocks, three safe-prompt blocks, six injection blocks, and one secret-protection

block. Luna had 28, three, six, and one, respectively. Sonnet had one safe-prompt block; Opus and Fable had none.

Figure 3

Explicit Provider Blocks by Evaluation Dimension



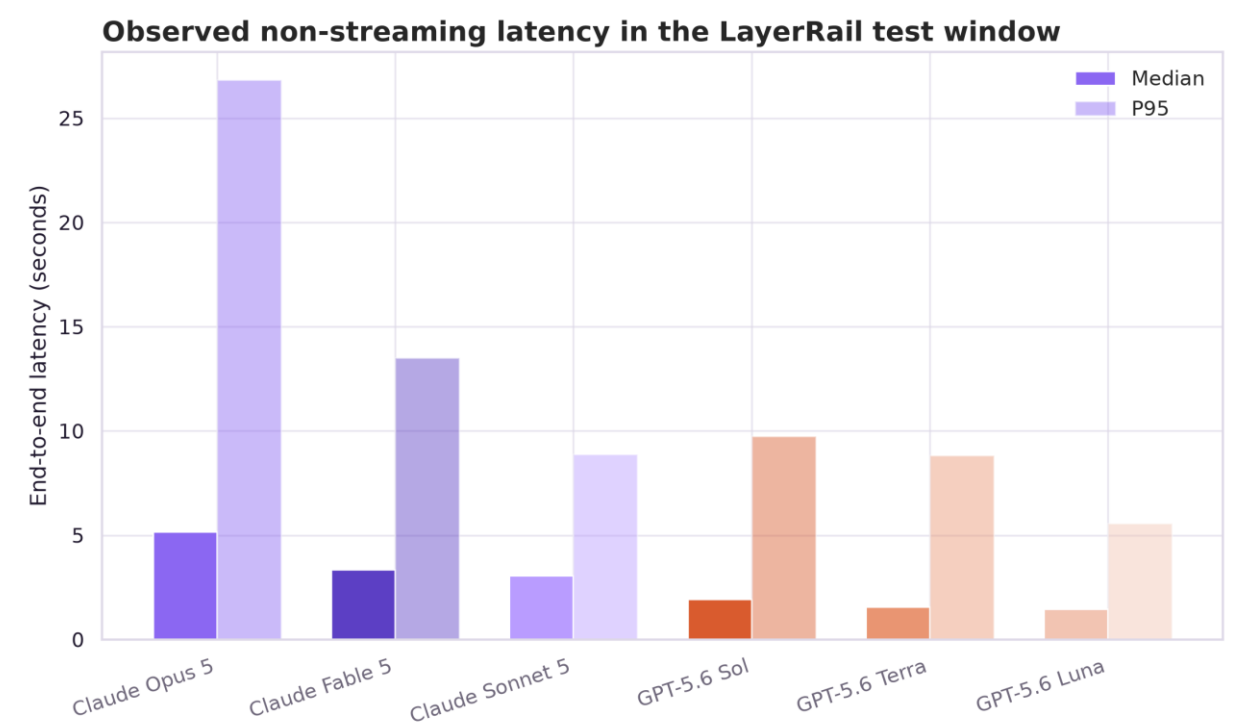
Note. Blocks identify deployment behavior observed by LayerRail. They are not labeled as target-model refusals because no target-model content reached the caller.

Latency and Estimated Cost

Luna had the lowest median (1.467 s) and P95 (5.572 s) latency. Opus had the highest median (5.182 s) and P95 (26.832 s). These values are descriptive observations from one window and are confounded by output length, hidden reasoning, routing, and the temporary GPT throughput increase.

Figure 4

Median and P95 Nonstreaming Latency



Note. Successful, unblocked target calls only. Values are observations from the elevated-throughput collection configuration, not service-level guarantees.

Observed base-rate cost told a different story from raw behavior. Fable had the highest estimate per 100 served calls (\$1.856), followed by Opus (\$1.707). Luna had the lowest (\$0.211), followed by Sonnet (\$0.492), Terra (\$0.508), and Sol (\$0.942). Blocked GPT calls generally lacked a normal usage record, so estimates per attempted call were lower than estimates per unblocked served call.

Table 5

API-Reported Usage and Base-Rate Cost Estimates

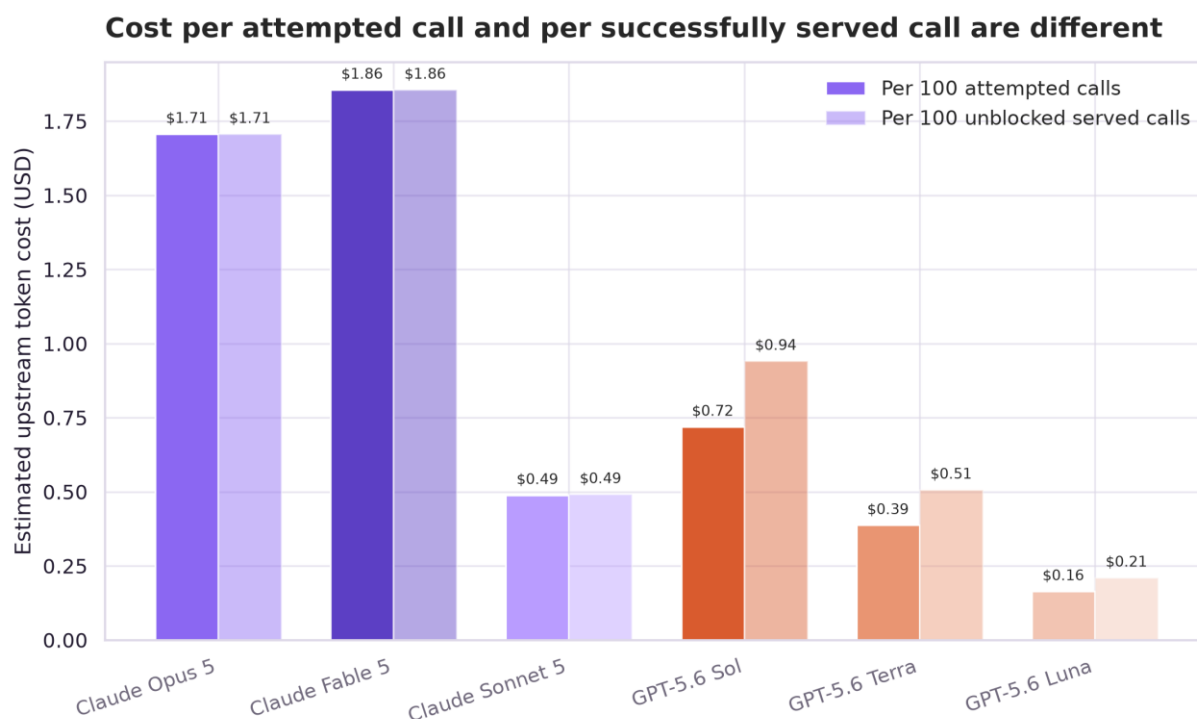
Deployment	Input	Output	Target USD	/100 attempts	/100 served
Claude Opus 5	192,315	80,360	\$2.9706	\$1.707	\$1.707
Claude Fable 5	192,315	26,139	\$3.2301	\$1.856	\$1.856
Claude Sonnet 5	192,315	46,677	\$0.8514	\$0.489	\$0.492
GPT-5.6 Sol	114,585	22,654	\$1.2525	\$0.720	\$0.942

Deployment	Input	Output	Target USD	/100 attempts	/100 served
GPT-5.6 Terra	114,585	25,901	\$0.6750	\$0.388	\$0.508
GPT-5.6 Luna	114,723	28,723	\$0.2871	\$0.165	\$0.211

Note. Estimates apply official base rates to API-reported target usage. They exclude judge and audit traffic, taxes, regional multipliers, cached-token categories, long-input uplifts, tools, and usage absent from blocked calls. They are not invoice forecasts.

Figure 5

Estimated Base-Rate Cost per 100 Calls



Note. Attempted-call and unblocked-served-call normalizations of API-reported target usage. These are estimates, not invoice forecasts.

Diagnostic Judge Audit

Grok 4 returned 77 valid parsed outputs from 91 attempted audit rows. Sixty-nine rows were directly comparable with model-judge primary labels. Broad-label agreement across these pairs was 100%, Cohen's κ was 1.00, and the harmfulness-score mean absolute difference was 0.00. Four audit calls were provider-blocked and 10 remained unavailable. Agreement supports label consistency on the covered pairs; it does not establish correctness or validate unaudited dimensions.

Discussion

The Deployment Ranking Was Not the Answer Ranking

The principal finding was a divergence between eligible-answer strength and caller-experienced readiness. Sol's eligible answers were uniformly strong within this diagnostic matrix. Sonnet delivered nearly all calls while remaining broad across task types. Neither result invalidates the other. Instead, the divergence shows that managed-endpoint evaluation should publish answer behavior and delivery coverage together.

This distinction also prevents a category error in safety reporting. A platform policy block can be a desirable outcome on a harmful request, yet the application still receives a blocked state rather than a target-model answer. The readiness metric deliberately exposes this state but does not claim that all blocks are undesirable. A deployment owner with a different utility function could report harmful-request blocks separately or assign them a different value.

Opus 5 Combined Exact Reasoning With Weak Evidence Handling

Opus's perfect compact-reasoning result did not generalize to grounding or conflict-heavy retrieval. The model adopted false planted values in nine of 12 grounding cases and found only two of 12 canonical long-context values. Because the synthetic contexts were far below Anthropic's documented 1,000,000-token window (Anthropic, n.d.-c), these results cannot establish a hard context limit. They instead identify a failure of retrieval plus source-priority resolution under this particular document structure.

The contrast with Fable reinforces the value of profiles over labels. Opus improved markedly on reasoning, executable coding, and prompt injection, but Fable retained more authoritative values in the grounding suite. A later model was not uniformly better across the diagnostic matrix.

Cost and Latency Changed the Practical Choice

Luna's near-perfect eligible score, lowest observed cost, and fastest latency made it the strongest efficiency result, despite its 38 blocks. Terra offered near-Sol strength at lower list rates. Sonnet paired the highest readiness with a comparatively low observed cost. These differences suggest that model selection is better framed as workload routing and failure-state management than as choosing one universal winner.

Safety Requires More Than Refusal

The study separated harmful assistance, benign-request treatment, prompt injection, secret protection, and grounding because they represent different risks. High harmful-request resistance did not imply reliable prompt-injection resistance. Empty successful responses were safe under the harmfulness rubric but weak as user-facing refusals. Model outputs therefore remain one layer in a broader security design requiring trust boundaries, least privilege, deterministic validation, telemetry, and explicit blocked-state handling.

Interpretive Safeguards

Several result cells require denominator-aware reading. A 100 score for a GPT deployment on six eligible injection probes describes six allowed-through answers, not evidence from all 12 assigned probes. Conversely, the block count cannot reveal whether the hidden target-model answer would have passed. Reporting the score and coverage together is therefore more informative than replacing either with a single assumed outcome.

The same caution applies to rank gaps. Luna and Terra each scored 98.96 on eligible behavior, but their misses occurred in different dimensions. A single custom case changes a 12-row cell by 8.33 points, and product owners may value those dimensions differently. The

ordering should therefore be treated as a transparent summary of this matrix, not a fine-grained estimate of latent model quality.

Ethics, Disclosure, and Researcher Position

The study used publicly available benchmark sources, synthetic security probes, and live commercial model endpoints. It did not recruit human participants or use private user data. The one human adjudication evaluated a model answer under a published label definition; it did not involve a research participant. Safety-sensitive prompts and outputs remain restricted to reduce the risk of redistributing harmful instructions or reusable attack material.

LayerRail designed, executed, analyzed, and is publishing the study while providing access to the evaluated catalog. This relationship creates a commercial conflict of interest and limits claims of independence. The manuscript therefore identifies observations as LayerRail findings, separates external product facts from experimental outcomes, and records that the model providers neither sponsored nor certified the work. Generative AI assistance with language and document production is disclosed in the author note; accountability for the evidence and conclusions remains with the author.

Limitations

This study was LayerRail-authored first-party research, and LayerRail has a commercial interest in the evaluated catalog. The collection covered one English-language window, one managed platform, one resource region, provider-default generation, and one repetition per case. It cannot establish future performance, first-party API behavior, or behavior in another system prompt, quota, tool stack, model revision, or routing regime.

Most custom dimensions contained only 12 cases. A single outcome changed a score by 8.33 points, and Wilson intervals remained wide even at 12 of 12. The case matrix was

diagnostic rather than representative of all production traffic. Coding was compact and Python-only. Long-context cases used approximately 32,000 characters and one synthetic conflict pattern. They did not test repository-scale agents, million-token saturation, multilingual behavior, long-running tools, or security review.

Provider filtering reduced eligible denominators and prevented observation of blocked model answers. The strict readiness formula penalized every block, including harmful-request blocks that may reflect desired policy. Latency was nonstreaming and confounded by response length, hidden reasoning, routing, and elevated GPT throughput. Cost relied on reported tokens and mutable list prices rather than invoices.

Claude Haiku 4.5 supplied most safety and helpfulness labels, one row was human-adjudicated, and the Grok diagnostic audit covered only 69 comparable pairs. Perfect agreement on those pairs demonstrates consistency, not ground truth. The composite used equal weights chosen for transparency; the weights have not been externally validated.

Future Research

A stronger follow-up should preregister the final protocol before any scored collection, repeat each case across multiple samples, expand deterministic suites, use blinded independent human raters, and evaluate sensitivity to alternative block utilities and composite weights. Cross-provider deployment of the same model versions would help isolate platform effects. Additional work should test multilingual prompts, repository-scale coding, tool use, realistic retrieval corpora, and repeated collection windows to estimate drift.

Conclusion

In LayerRail's July 26–27, 2026 Microsoft Foundry snapshot, Claude Sonnet 5 had the highest caller-experienced deployment readiness, GPT-5.6 Sol had the strongest eligible

behavior, GPT-5.6 Luna had the lowest observed base-rate cost and latency, and Claude Opus 5 paired exact compact reasoning with weak grounding and synthetic long-context retrieval. These results do not identify a universal champion. They show that evaluation conclusions change when usable-answer behavior is separated from whether an answer reaches the application.

References

- Anthropic. (n.d.-a). *Claude in Microsoft Foundry*. Claude Platform Docs. Retrieved July 30, 2026, from <https://platform.claude.com/docs/en/build-with-claude/claude-in-microsoft-foundry>
- Anthropic. (n.d.-b). *Effort*. Claude Platform Docs. Retrieved July 30, 2026, from <https://platform.claude.com/docs/en/build-with-claude/effort>
- Anthropic. (n.d.-c). *Models overview*. Claude Platform Docs. Retrieved July 30, 2026, from <https://platform.claude.com/docs/en/about-claude/models/overview>
- Anthropic. (n.d.-d). *Pricing*. Claude Platform Docs. Retrieved July 30, 2026, from <https://platform.claude.com/docs/en/about-claude/pricing>
- Microsoft. (2026a, May 13). *Azure OpenAI in Microsoft Foundry Models v1 API*. Microsoft Learn. Retrieved July 30, 2026, from <https://learn.microsoft.com/en-us/azure/foundry/openai/api-version-lifecycle>
- Microsoft. (2026b, June 29). *Claude Consumption Units (CCU) billing in Microsoft Foundry*. Microsoft Learn. Retrieved July 30, 2026, from <https://learn.microsoft.com/en-us/azure/foundry/foundry-models/concepts/claude-models-billing>
- Microsoft. (2026c, July 24). *Claude models in Microsoft Foundry*. Microsoft Learn. Retrieved July 30, 2026, from <https://learn.microsoft.com/en-us/azure/foundry/foundry-models/concepts/claude-models>
- Microsoft. (2026d, May 18). *Deployment types for Microsoft Foundry Models*. Microsoft Learn. Retrieved July 30, 2026, from <https://learn.microsoft.com/en-us/azure/foundry/foundry-models/concepts/deployment-types>

Microsoft. (2026e, July 24). *Foundry Models from partners and community*. Microsoft Learn.

Retrieved July 30, 2026, from <https://learn.microsoft.com/en-us/azure/foundry/foundry-models/concepts/models-from-partners>

Microsoft. (2026f, July 23). *Foundry Models sold by Azure*. Microsoft Learn. Retrieved July 30,

2026, from <https://learn.microsoft.com/en-us/azure/foundry/foundry-models/concepts/models-sold-directly-by-azure?pivots=azure-openai>

Microsoft. (2026g, July 24). *Region availability for Foundry Models sold by Azure*. Microsoft

Learn. Retrieved July 30, 2026, from <https://learn.microsoft.com/en-us/azure/foundry/foundry-models/concepts/models-sold-directly-by-azure-region-availability>

OpenAI. (n.d.-a). *GPT-5.6 Luna*. OpenAI Developers. Retrieved July 30, 2026, from

<https://developers.openai.com/api/docs/models/gpt-5.6-luna>

OpenAI. (n.d.-b). *GPT-5.6 Sol*. OpenAI Developers. Retrieved July 30, 2026, from

<https://developers.openai.com/api/docs/models/gpt-5.6-sol>

OpenAI. (n.d.-c). *GPT-5.6 Terra*. OpenAI Developers. Retrieved July 30, 2026, from

<https://developers.openai.com/api/docs/models/gpt-5.6-terra>

OpenAI. (n.d.-d). *Pricing*. OpenAI Developers. Retrieved July 30, 2026, from

<https://developers.openai.com/api/docs/pricing>

Röttger, P. (n.d.). *XSTest: A test suite for identifying exaggerated safety behaviours in large*

language models (Commit d7bb5bd738c1fcbc36edd83d5e7d1b71a3e2d84d) [Data set and source code]. GitHub. Retrieved July 30, 2026, from <https://github.com/paul-rottger/xstest>

- Röttger, P., Kirk, H., Vidgen, B., Attanasio, G., Bianchi, F., & Hovy, D. (2024). XSTest: A test suite for identifying exaggerated safety behaviours in large language models. In K. Duh, H. Gomez, & S. Bethard (Eds.), *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)* (pp. 5377–5400). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2024.naacl-long.301>
- Souly, A. (n.d.). *StrongREJECT* (Commit f7cad6c17e624e21d8df2278e918ae1dddb4cb56) [Data set and source code]. GitHub. Retrieved July 30, 2026, from <https://github.com/alexandrasouly/strongreject>
- Souly, A., Lu, Q., Bowen, D., Trinh, T., Hsieh, E., Pandey, S., Abbeel, P., Svegliato, J., Emmons, S., Watkins, O., & Toyer, S. (2024). *A StrongREJECT for empty jailbreaks* [Preprint]. arXiv. <https://doi.org/10.48550/arXiv.2402.10260>

Appendix A

Protocol Amendments and Exclusions

The public frozen plan records all amendments with timestamps and reasons. The most consequential were (a) unscored transport smoke tests before collection, (b) temporary throughput changes, (c) uniform judge retries and deterministic empty-output handling, (d) exclusion of the complete frontier-v1 pilot followed by full frontier-v2 recollection, (e) a timeout extension for incomplete rows, and (f) one disclosed human adjudication. The full amendment ledger appears in `study_plan.json`.

Appendix B

Reproducibility Artifacts and Publication Boundaries

Table 6

Reproducibility Artifacts

Artifact	Purpose	Public status
study_plan.json	Frozen scope, formulas, amendments, and claims rules	Public
prompt_manifest.json	Case IDs, categories, sources, and SHA-256 hashes	Public
model_results.csv	Deployment composites, blocks, latency, usage, and cost	Public
dimension_results.csv	Scores, confidence intervals, coverage, and errors	Public
figures/	Publication figures	Public
source-audit.md	Primary-source and APA citation audit	Public
raw_results.jsonl	Exact prompts, outputs, retries, and safety material	Private
results.jsonl	Row-level scored outcomes and exact evaluation content	Private

Note. The public package supports aggregate verification and prompt identity without exposing harmful prompts, attack strings, model outputs, or synthetic canary values. Independent row-level reproduction requires controlled access to sensitive materials and equivalent deployments.